Project Management Plan (PMP)

Project Title: Hear My Hands

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Project Overview

Hear My Hands is an AI-based system that translates American Sign Language (ASL) hand gestures into text and speech using computer vision and deep learning. The technology employs deep learning models to precisely identify gestures, such as alphabets and frequently used greetings, and leverages sophisticated computer vision algorithms to record hand movements through a camera.

After a gesture is identified, it is processed and turned into readable text. A text-to-speech engine then converts that text into audible speech. This makes it possible for people who use sign language and others who don't to communicate easily.

Hear My Hands' main objective is to improve accessibility, efficiency, and inclusivity by bridging the communication gap between the deaf or mute communities and the broader public.

Objectives:

- Recognize A–Z ASL alphabets.
- Detect 20 common ASL greetings.
- Convert gestures into text.
- Convert text into speech (TTS)
- Provide real-time communication support.

Project Scope

In Scope:

- Hand gesture detection using camera.

- Training dataset for ASL alphabets & greetings
- AI model using TensorFlow.
- Real-time prediction system
- Text-to-Speech output
- Simple user interface

Out of Scope:

- Full sentence translation (beyond trained greetings)
- Mobile app development
- Multi-language sign support
- Hardware device integration

Project Cost

Category	Cost (AUD)
Software (Python, OpenCV, TensorFlow)	Free (Open Source)
Hardware (Webcam, PC)	$200
Storage	Local (Free)
Miscellaneous	$100
Total Estimated Cost	$300 AUD

Project Timeline (12 Weeks)

Here's the attached Gantt chart to provide a idea about our project Timeline.

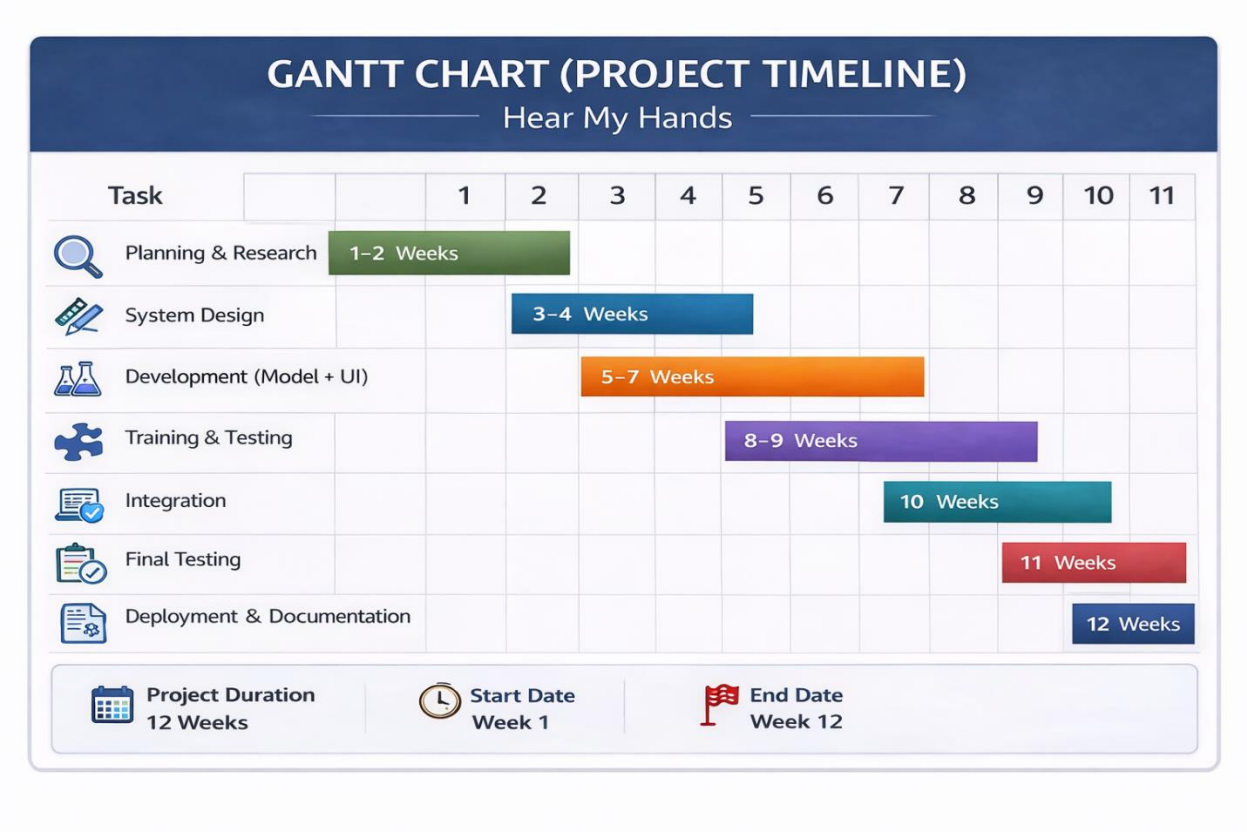


Figure 1: Gantt Chart

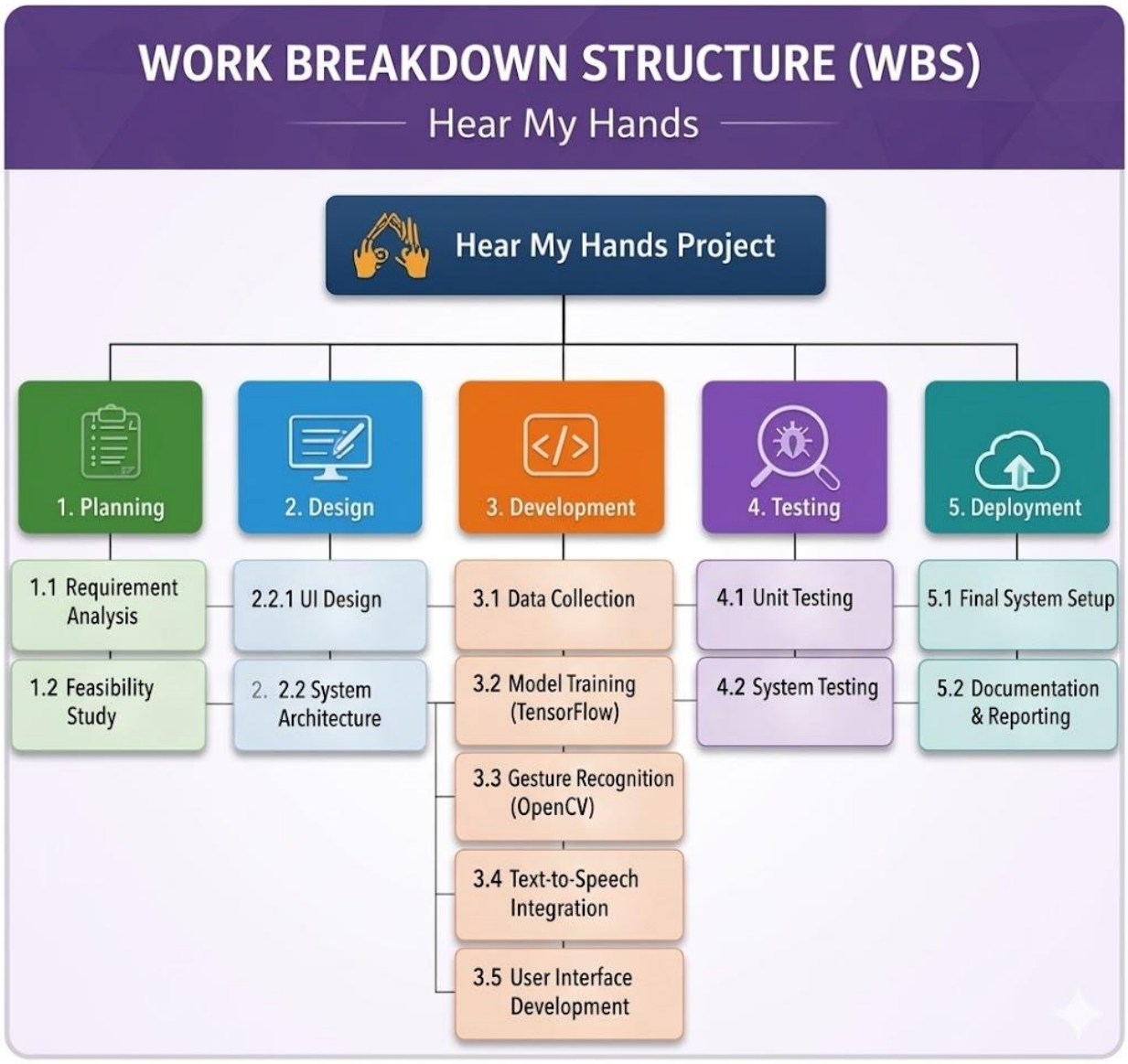


Figure 2: Work Breakdown Structure (WBS)

Risk Management

Risk	Impact	Probability	Mitigation
Poor model accuracy	High	Medium	Use more training data

Lighting issues	Medium	High	Improve preprocessing
Hardware limitations	Medium	Medium	Optimize code
Time delays	High	Medium	Follow schedule strictly
Integration errors	Medium	Medium	Test modules separately

Milestones

Milestone	Week
Project Approval	Week 3
Design Completed	Week 5
Model Developed	Week 9
Testing Completed	Week 11
Final Submission	Week 12

Tools & Technologies

The tools and technologies we will be using for this project are as follows:

- Python
- OpenCV
- TensorFlow
- NumPy
- Text-to-Speech (pyttsx3 / gTTS)

Quality Management

- Ensure model accuracy ≥ 85%.

- Test in different lighting conditions
- Validate real-time performance.
- Ensure clear speech output.

Communication Plan

Stakeholder	Communication Method	Frequency
Supervisor (Professor)	Reports/Meetings	Weekly
Team members	Chat/Docs	Daily

Deliverables

- Hear my Hands Model
- Python Application
- Documentation Report
- Presentation (PPT)
- Demo System

Conclusion:

The Hear My Hands project aims to bridge the communication gap between individuals who use sign language and those who do not, by leveraging the power of artificial intelligence and computer vision. By translating hand gestures into text and speech in real time, the system provides an inclusive and accessible solution that can significantly improve everyday communication for the deaf and mute community.

With a well-defined project scope, clear objectives, and a structured development approach, our project ensures that each phase: from planning and design to development,

testing, and deployment is carefully managed. Effective risk management strategies have been considered to address potential challenges such as model accuracy, environmental conditions, and system integration issues.

Overall, with proper planning, continuous monitoring, and systematic execution, the Hear My Hands project is both practical and achievable within the proposed 12-week timeline. Beyond its technical implementation, the project also contributes to social impact by promoting inclusivity, accessibility, and improved human interaction through innovative technology.